

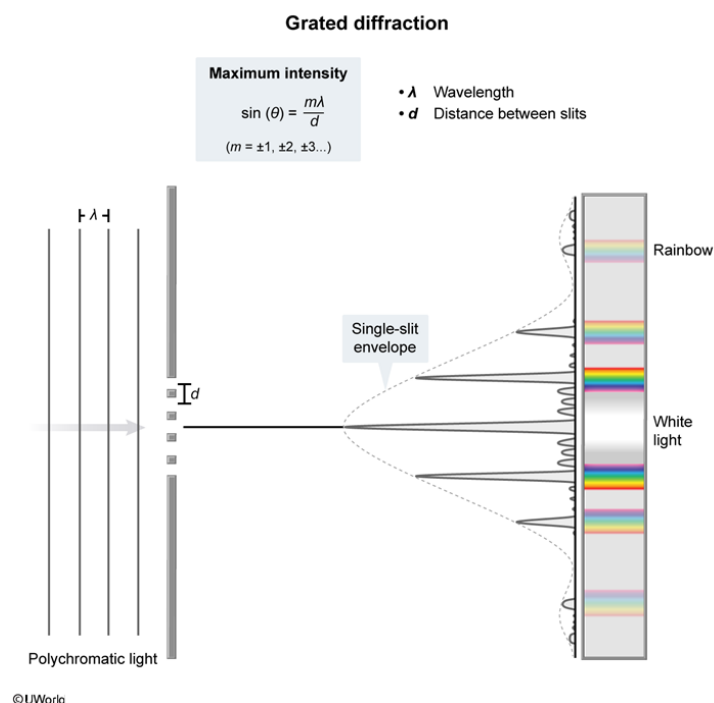
Which of the following procedures best describes the use of diffractive phenomena in separating the individual components of an electromagnetic wavefront?

- ☐ A. Single-slit diffraction of monochromatic light [19%]
- ☐ B. Double-slit diffraction of polychromatic light [28%]
- ✓ ☒ C. Grated diffraction of polychromatic light [39%]
- ☐ D. Grated diffraction of monochromatic light [12%]

Correct

38% answered correctly

Explanation:



A **diffraction grating** is a diffractive array composed of **numerous slits** with near-zero width similar to the wavelength of incident light. Like the band patterns produced through [single-slit diffraction](#) and [double-slit diffraction](#), dark bands that result from light passing through a diffraction grating are caused by [destructive interference](#) between light waves. Conversely, light bands are caused by constructive interference between light waves.

Compared to other diffractive phenomena, the band pattern caused by

grated diffraction is characterized by relatively **sharp and narrow peaks** (ie, thin peaks with highly variable amplitude). This band pattern occurs because constructive and destructive interference occurs not only between waves emerging from adjacent slits (eg, double-slit diffraction) but also between waves emerging from slits separated from each other by other slits.

Light bands (regions of maximum light intensity) from the diffraction grating occur when diffracted light waves pass through the slits at angles (θ) that are a function of the distance separating each slit (d) and an integer multiple ($m = \pm 1, \pm 2, \pm 3 \dots$) of the wavelength:

$$\sin \theta = \frac{m \lambda}{d}$$

The sharp, narrow peaks of the band pattern produced by a diffraction grating are useful in analyzing the properties of the light that enters the slit. For example, **long-wavelength light** diffracts more than short-wavelength light. Consequently, transmitting **polychromatic light** composed of light waves with different wavelengths (eg, white light) through a diffraction grating will produce a series of multicolored arrays (eg, rainbows) that enables an analysis of wavefront composition.

(Choices A and B) Compared to grating diffraction, the band patterns produced by single- and double-slit diffraction are relatively diffuse. Therefore, single- and double-slit diffractions are not helpful in analyzing the individual components of light wavefronts.

(Choice D) In contrast to polychromatic white light, monochromatic light is composed of only one color (ie, one wavelength). Consequently, the wavelength of monochromatic light can be detected without the use of diffraction.

Educational objective:

Diffraction grating refers to passage of light through an array consisting of numerous slits. Diffraction grating may be used to determine the components of polychromatic (eg, white) light and other mixed electromagnetic wavefronts.

Time spent:0 sec
Subject:
Physics

QID:402433
System:
Light and Sound

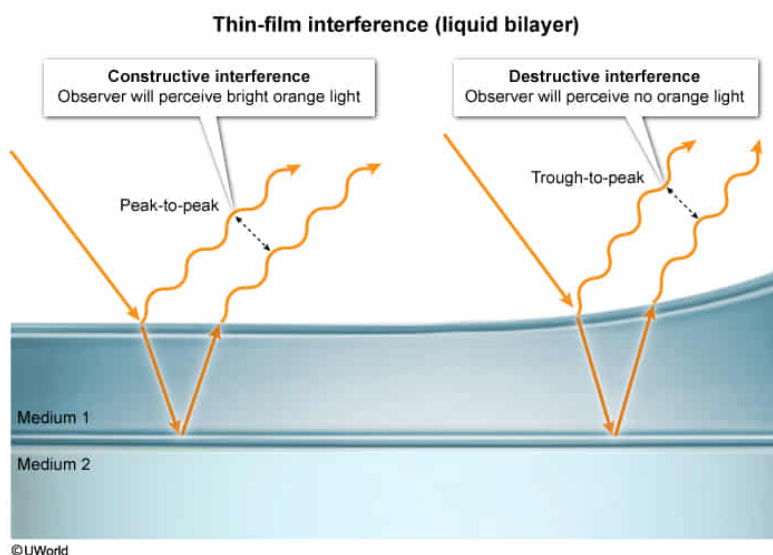
Thin-film interference is observed when polychromatic light is incident on an interface formed when one semitransparent medium is layered on top of a second semitransparent medium. Which of the following best explains why thin-film interference generates a multicolored array when two semitransparent fluids are used?

- ☐ A. The density of the top fluid varies along the fluid interface. [23%]
- ✓ ☒ B. The thickness of the top fluid varies along the fluid interface. [24%]
- ☐ C. The osmolarity of the top fluid varies along the fluid interface. [5%]
- ☐ D. The two semitransparent fluids decrease the number of reflection events at the fluid interface. [46%]

Correct

24% answered correctly

Explanation:



Thin-film interference refers to the **multicolored arrays** generated by the reflection events that occur within a system composed of

two layers of **semitransparent media**. The relative intensity of the **colors** seen within the multicolored array depends on the constructive and destructive **interference** among waveforms of **polychromatic** (variable wavelength) light.

Interference that occurs within a thin-film system results from differences in the travel path of the light. Some light that is incident on the two-media system will reflect off the surface of the first medium (thin film), and some light will transmit through the first medium before reflecting off the media interface.

When light reflected off the surface of the first medium constructively interferes with light reflected off the media interface, an observer perceives bright light. Conversely, when light reflects off the surface of the first medium and destructively interferes with light reflected off the media interface, an observer perceives darkness.

Thin-film interference is influenced by the **thickness** of the film because light waves of a specific wavelength (ie, color) that interfere constructively or destructively at one film thickness may no longer interfere in the same fashion at a different thickness. Therefore, **surface tension** and other effects that locally distort the surface of a fluid may also cause multicolored arrays to appear on the surface as light that emerges from adjacent segments of the thin film varies in color composition.

(Choices A and C) Variations in fluid density and osmolarity cannot explain changes to the interference pattern of light that are necessary to produce multicolored arrays. Furthermore, the fluid density and osmolarity of each medium would not be expected to vary within the same two-media system.

(Choice D) Layering two semitransparent fluids *increases* the number of reflection events, enabling the waveform interference

that generates multicolored arrays.

Educational objective:

Thin-film interference describes the multicolored array generated when polychromatic light is incident on an interface formed by two semitransparent media. Localized discrepancies within the thickness of a thin film (top media) contribute to variable interference patterns that produce the colorful array.

Time spent:0 sec

Subject:
Physics

QID:402434

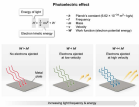
System:
Light and Sound

The photoelectric effect refers to the ejection of electrons from a material when the material absorbs electromagnetic radiation of sufficient energy. As the frequency of electromagnetic radiation increases, what happens to the kinetic energy of ejected electrons?

- ☐ A. It decreases. [5%]
- ☐ B. It remains constant. [11%]
- ☒ C. It increases. [79%]
- ☐ D. It changes but is not dependent on the energy of electromagnetic radiation. [3%]

Correct
79% answered correctly

Explanation:



The **photoelectric effect** describes the ejection of electrons from a substance due to the **absorption of electromagnetic radiation**. Although higher-intensity electromagnetic radiation exhibits greater **electric field oscillations**, some types of electromagnetic radiation do not cause the ejection of electrons in any case, regardless of intensity. This is due to the **electric potential energy** between positive charges (protons) and negative charges (electrons).

For each electron there exists a discrete value of electric potential known as the **work function (W)**, and the magnitude of energy absorbed through electromagnetic radiation must exceed W for the ejection of an electron to occur.

The energy of electromagnetic radiation (E) is proportional to its frequency (f) via Planck's constant (h = 6.62 × 10⁻³⁴ m² kg/s), as given by:

$$E = hf$$

Accordingly, the ejection of an electron from a surface by incident electromagnetic radiation depends on the frequency of incident electromagnetic radiation. Furthermore, the conservation of energy dictates that the energy absorbed by an electron must contribute to either the ejection of the electron (overcoming the work function) or to the **kinetic energy** of the electron following ejection:

$$hf = \frac{1}{2}mv^2 + W$$

Consequently, increasing the energy of electromagnetic radiation (ie, frequency) beyond the value of the work function will increase the kinetic energy of ejected electrons.

(Choices A and B) Increasing the energy of electromagnetic radiation *must* increase the kinetic energy of ejected electrons because the work function of each electron is fixed, and the conservation of energy must be maintained.

(Choice D) The kinetic energy of ejected electrons must obey the conservation of energy. Absorbed energy from electromagnetic radiation that does not contribute to surpassing the work function is converted to the kinetic energy of ejected electrons.

Educational objective:

The photoelectric effect describes the ejection of electrons from a substance due to the absorption of electromagnetic radiation with sufficient energy. Because the energy needed to eject an electron is fixed via its work function, increasing the energy of electromagnetic radiation will increase the kinetic energy of ejected electrons.

Time spent: 0 sec
Subject: Physics

QID: A02435
System: Light and Sound